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Short communication

Study of anisotropy impact of 2A12 and 7075 dissimilar aluminum alloys FSW joints

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ABSTRACT

The influence of anisotropy on dissimilar aluminum alloy FSW joints were studied by changing the welding direction and advancing side materials. The results show the significant differences in both mechanical properties and microstructure of the welded joints. The 0°7AS joint has the highest tensile strength (414 MPa), reaching 87 % of the 2A12 aluminum alloys strength. The 0°7AS joint has the smallest average grain size in the weld nugget, and the best recovery and recrystallization ability. In the HAZ, the strongest texture types of the 0°7AS and 45°7AS joints are different. The 0°7AS joint has the highest texture strength, with a prominent {011}<211> brass texture. This paper provides the guidance for aluminum alloys welding.

1. Introduction

Friction Stir Welding (FSW), as a solid-state joining technique, is one of the ideal craft for welding aluminum alloys, effectively mitigating the joint defects commonly associated with traditional fusion welding methods [1–4]. The effect of speed, welding speed, stir-welding head pressure, and shoulder pressure on dissimilar aluminum alloys FSW joints were studied [5–9].

Cetkin [5] conducted FSW on AA7075 and AA5182 plates with conical stri-welding head, and the results showed that when the rotating speed was 980 rpm and the welding speed was 108 mm/min, the two aluminum alloys in the FSW joint NZ were mixed evenly, and the tensile strength of the joint can reach 265 MPa. The FSW joint of AA2219-O and AA 7475-T73 plates were investigated by Khan [6]. When the rotating speed and the welding speed were 900 rpm and 100 mm/min, respectively, there were no obvious defects in the cross section of the joint, and the tensile strength of the joint is 267 MPa, reaching 92 % of the tensile strength of AA2219-O. Saravanan [7] found that the tensile strength of the FSW 2024/7075 joint can reach 356 MPa, when the ratio of axle shoulder diameter to cylindrical stirring needle diameter was 3:1. Kalinenko [9] showed that the FSW 6061/7075 joint has a good welding quality under the shoulder pressure with 6.7 KN. In the above research, these aluminum alloy plates are usually prepared by rolling. After rolling, the mechanical properties of each direction of these plates will be obviously different [10,11]. However, little researches pay attention on anisotropy on mechanical properties of FSW joints.

Therefore, the dissimilar 2A12 and 7075 aluminum alloy FSW joints were obtained in this paper,and the effect of anisotropy on mechanical properties of FSW joints was investigated, which will provide guidance for aluminum alloy welding.

2. Materials and methods

The chemical compositions of the experimental materials were shown in Table 1.

FSW experiments were carried on by the FSW-LM-BM16 CNC friction stir welding machine, and welding parameters was 1000r/min of stir-welding head speed and 80 mm/min of welding speed. The stir-welding head was H13 steel, and the pin was a tapered threaded with a shoulder diameter of 12 mm and pin length of 4.6 mm, with stir-welding head tilt angle of 2.5°. The samples are named as 0°7AS, 45°7AS, 90°7AS, 0°2AS, 45°2AS, 90°2AS by the plate welding direction and advancing side material. Tensile tests were conducted using the CMT-5205 universal testing machine for more than three times. The gauge section of the tensile specimen is 5.0 × 2.5 mm with the gauge length of 30.0 mm. Metallographic specimens were prepared using an EDM wire cutting machine and the samples were characterized by electron backscatter diffraction (EBSD) using an electrolyte of 10 % $HClO_4$ solution and 90 % anhydrous ethanol under a polishing process with a voltage of 20 V and a polishing time of 15 s.

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Table 1
Chemical composition of 2A12-T4 and 7075-T6 (wt. %).

Material	Si	Fe	Cu	Mn	Mg	Ni	Zn	Ti	Cr	Al
2A12	0.50	0.50	3.8–4.9	0.30–0.90	1.2–1.8	0.10	0.30	0.15	–	Bal
7075	0.40	0.50	1.2–2.0	0.30	2.1–2.9	–	5.1–6.1	0.20	0.18–0.28	Bal

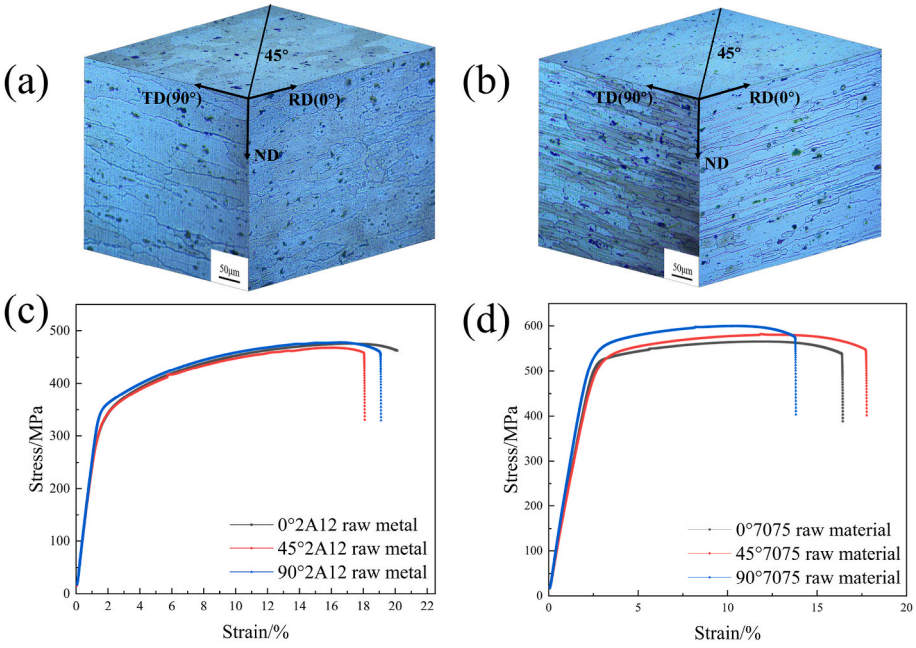


Fig. 1. (a)–(b) Microstructure and (c)–(d) tensile properties of 2A12 and 7075 aluminum alloys.

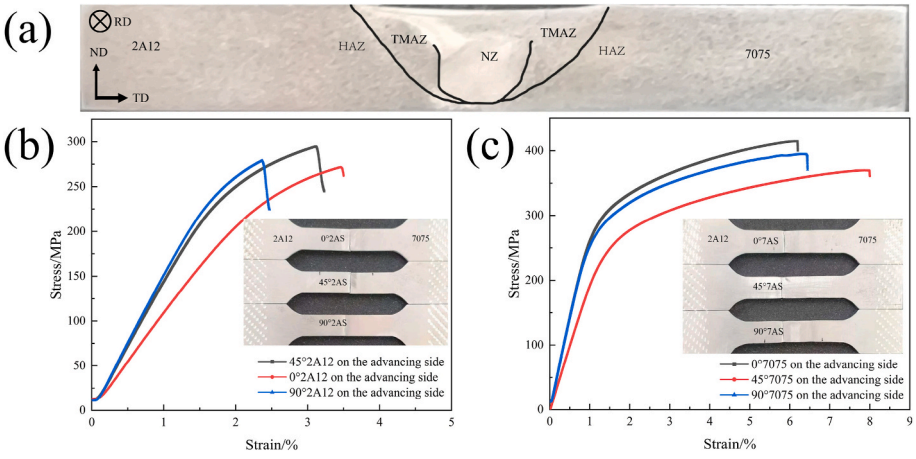


Fig. 2. (a) Macroscopic morphology of 0°7AS joints and (b)–(c) tensile properties of 2AS joints and 7AS joints.

3. Results and discussion

3.1. Raw material analysis

The microstructure of 2A12 and 7075 aluminum alloy is shown in Fig. 1(a)–(b), and the raw materials are typical rolled microstructure. There are the significant differences in grain morphology and size along the 0°(RD), 90°(TD), and normal directions (ND). The grains of 7075 aluminum alloy are finer along the 0° and 90° directions compared to the normal direction. The tensile results of the 2A12 and 7075 aluminum alloys are presented in Fig. 1(c)–(d). The tensile strength of the 2A12 alloys show the little variation at three directions. However, the tensile strength of 7075 aluminum alloy in 0°, 45° and 90° directions are 565 MPa, 582 MPa, 599 MPa, which is due to that the grain morphology of 7075 aluminum alloy in different directions are obviously different. Anisotropy can be expressed using the following formula [12]:

$$IPA = \frac{2X_{max} - X_{min} - X_{mid}}{2X_{max}} \times 100\% \quad (1)$$

Where X_{max}, X_{mid} and X_{min} represent the maximum, intermediate and minimum values of mechanical properties, respectively. The IPA values of 2A12 and 7075 aluminum alloys are 1.2 % and 4.2 %, respectively. The higher IPA value indicates the stronger anisotropy of the material.

Table 2
Mechanical properties of FSW joints.

sample	δ_b/-MPa	δ/%	sample	δ_b/-MPa	δ/%
0°7AS	414	6.2	0°2AS	292	3.5
45°7AS	369	8.0	45°2AS	294	3.2
90°7AS	393	6.6	90°2AS	279	2.5
IPA/%	7.9	20	IPA/%	2.8	18.5

3.2. Welded joints analysis

The macrostructure of the welded joints is shown in Fig. 2(a), the boundaries between the weld nugget zone (NZ), thermomechanical affected zone (TMAZ), and heat affected zone (HAZ) are clearly visible. The tensile properties of the joints are presented in Fig. 2(b)–(c) and Table 2, the tensile strength and elongation of the 7AS joints are both higher than those of the 2AS joints. The 7AS joints fracture in the HAZ on the 2A12 aluminum alloy side, while the 2AS joints fracture in the NZ. The melting point of 2A12 aluminum alloys is 560 °C, which is lower than 7075 aluminum alloys melting point (635 °C). Meanwhile, due to the effect of friction, the temperature of the advancing side is higher than that of the retreating side [13]. Therefore, the welding temperature of these 2AS joints is much lower than the melting point of 7075 aluminum alloy, which will lead to poor plastic fluidity of 7075 aluminum alloy in 2AS joints. As a result, these two aluminum alloys of 2AS joints do not mix evenly in the NZ, which cause the fracture to occur. Therefore, the mechanical properties of 7AS joints are better than those 2AS joints. The 0°7AS joint has the highest tensile strength (414 MPa), reaching 87 % of the strength of 2A12 aluminum alloy. Meanwhile, the calculated anisotropy index of the 7AS joint is 7.9 %.

The performance of the welded joint mainly depends on their microstructure. Fig. 3 shows the microstructure of the NZ in the 7AS joints. Fig. 3(b) illustrates the distribution of W_{HAGBs} and W_{LAGBs} as the welding direction changes from 0° to 90°. The proportion of W_{LAGBs} increases from 10 % to 14.1 % and then decreases to 11.9 % as the welding direction transitions from 0° to 90°. This is mainly because when welding along the 0° direction, the material exhibits better flowability and undergoes larger plastic deformation [4], which increases the ability of W_{LAGBs} to absorb dislocations and transform into W_{HAGBs}, leading to dynamic recrystallization and the formation of finer equiaxed grains [14,15]. Fig. 3(c) presents the grain structure characteristics of the NZ regions for the three joints. The 0°7AS joint exhibits large recrystallized grains in the NZ, while the 45°7AS and 90°7AS joints show relatively the fewer recrystallized grains. As the welding direction changes from 0° to 90°, the proportion of recrystallized grains in the weld nugget decreases. Based on the metallographic microstructure of the 2A12 and 7075 aluminum alloys in Fig. 1(a)–(b), during welding, the stirring action of the tool is likely to fragment the grains, promoting recovery and recrystallization. In contrast, the larger grains in the other two directions hinder the expansion of substructure grains, thus impeding recrystallization [15]. As shown in Fig. 3(a), The average

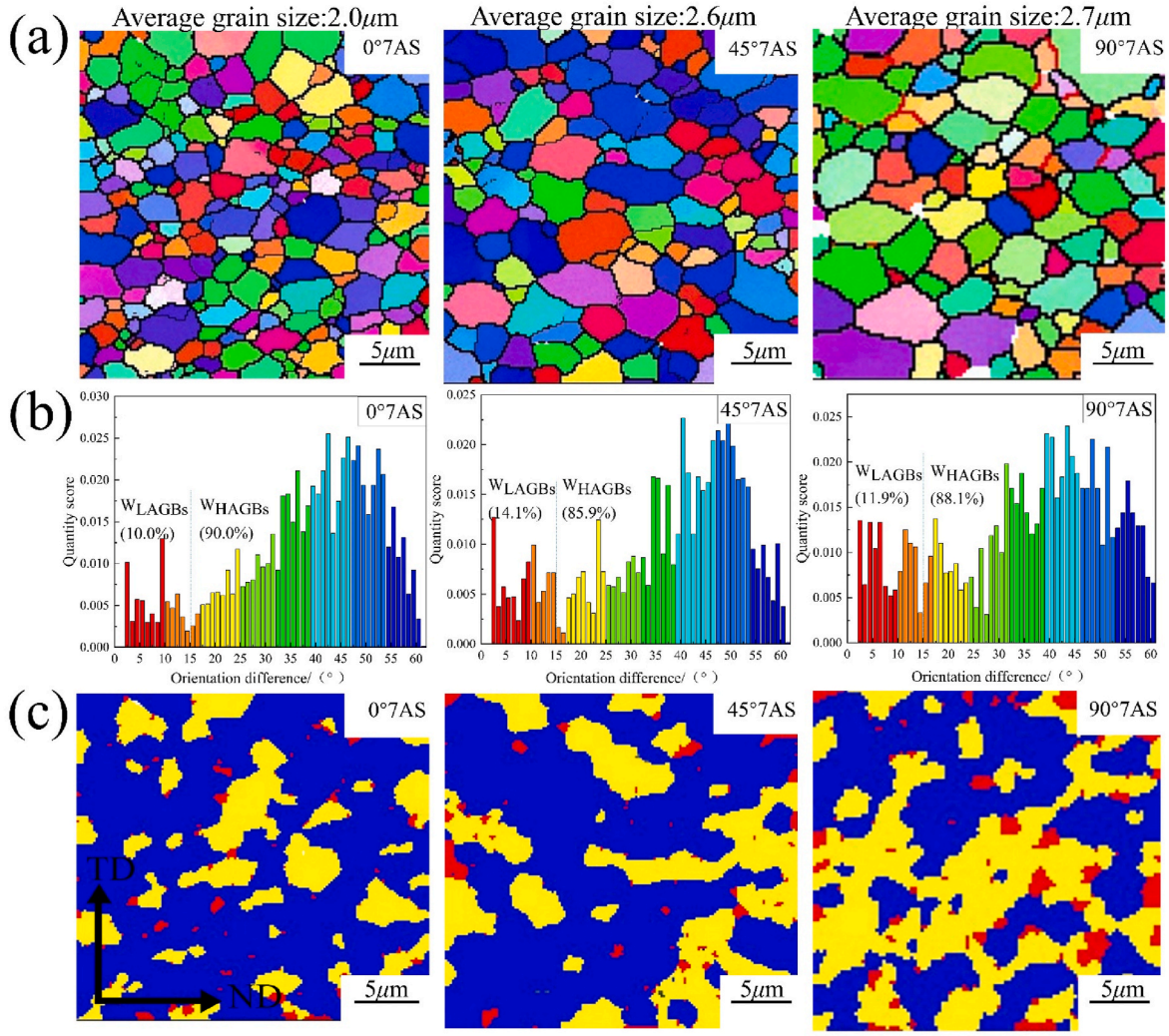


Fig. 3. (a) Average grain size, (b) W_{HAGBs} and W_{LAGBs} and (c) grain microstructure features of 7AS joints.

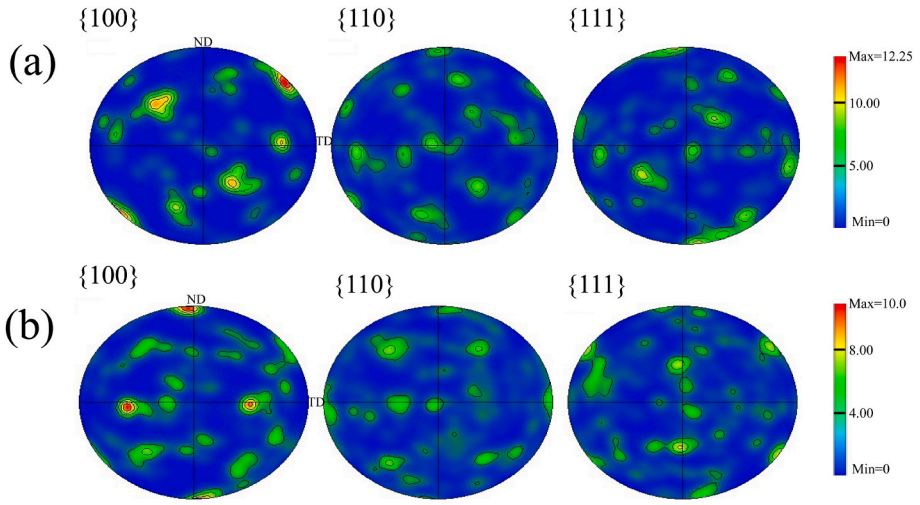


Fig. 4. Polar figures of the HAZ of the(a) 0°7AS and (b) 45°7AS joint.

grain sizes in the weld nugget zone for the 0°7AS, 45°7AS, and 90°7AS joints are 2.0 μm, 2.6 μm, and 2.7 μm. The results showed that the 0°7AS joint had the best grain refinement effect.

HAZ is a critical factor influencing the breaking of the joint. The breaking of 7AS joint is mainly concentrated in the HAZ. The tensile strength was the highest in the 0° direction and the lowest in the 45° direction. Fig. 4(a)–(b) is the polar figures of the HAZ in the direction of 0° and 45°, and the texture can be clearly seen. In the FSW process, the HAZ is primarily affected by thermal cycling, which causes grain growth and deformation, resulting in the formation of deformation textures [18]. In the 0°7AS joint, the strongest texture in the HAZ is the {011} <211> brass texture, and the highest strength is 12.25. For the 45°7AS joint, the strongest texture in the HAZ is the {001} <110> cubic texture, and the highest strength is 10, while a small amount of the {011} <211> brass texture is also present. The {011} <211>brass texture has the effect of reducing the strength in the 45° direction, and {001} <110>cubic texture has the same strength in all directions, but has the highest elongation due to the small amount of {011} <211>brass texture [16–18].

4. Conclusion

The effect of anisotropy on dissimilar aluminum alloy FSW joints of 2A12-T4 and 7075-T6 aluminum alloys was investigated. The main conclusions are as follows:

(1) The anisotropy index of the 7AS joint is 7.9 %, and the 0°7AS joint has the highest tensile strength (414 MPa), reaching 87 % of the 2A12 aluminum alloy strength.
(2) The 0°7AS joint has the smallest average grain size with 2.0 μm in the NZ. This is mainly because when welding along the 0° direction, the material exhibits better flowability and undergoes the larger plastic deformation, which is beneficial to improve the recrystallization ability.
(3) In the HAZ, the strongest texture types of the 0°7AS and 45°7AS joints are different. The 0°7AS joint has the highest texture strength, with a prominent {011} <211> brass texture.

CRediT authorship contribution statement

Yongqi Wang: Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Formal analysis, Data curation. **Zhihao Bai:** Writing – review & editing, Project administration, Investigation, Data curation. **Pengyu Wang:** Writing – review & editing, Investigation, Data curation. **Wanting Du:** Writing – review & editing, Investigation, Data curation. **Rui Huang:** Writing – review & editing, Investigation, Data curation. **Zhengwei Tang:** Writing – original draft, Investigation, Data curation. **Jiangnan Xia:** Investigation, Data curation. **Dongshuai Zhou:** Writing – review & editing, Formal analysis. **Jian Wang:** Validation, Methodology, Investigation.

Originality statement

This paper has not been published or presented elsewhere in part or in entirety and is not under consideration by another journal.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work.

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Data availability

Data will be made available on request.

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